INCH-POUND

MIL-DTL-85052/10A
29 December 2016
SUPERSEDING
MIL-DTL-85052/10
1 December 2015

# DETAIL SPECIFICATION SHEET

## CLAMP, LOOP, ALUMINUM 2024-T42, 275 °F, NITRILE BUTADIENE RUBBER FUEL AND PETROLEUM BASED HYDRAULIC FLUID RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-85052.

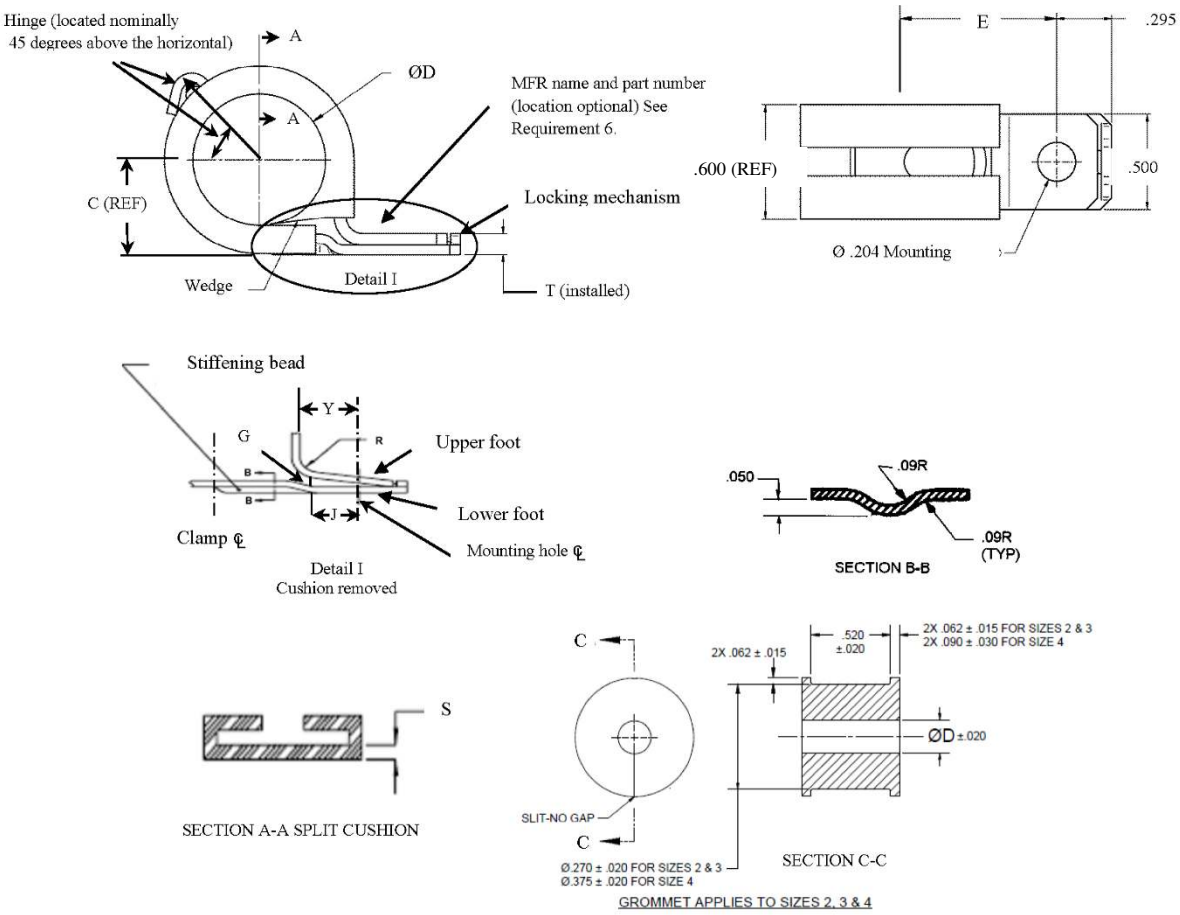



FIGURE 1. Clamp design and construction.



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TABLE I. Clamp dimensions.

| Dash No. | Harness Nominal O.D. 5/ | C REF | ØD 1/ 5/ | E 2/ 6/ ± .032 | G | J 4/ REF | R ± .010 | S ± .010 | T ± .010 | Y 4/ REF |
|---|---|---|---|---|---|---|---|---|---|---|
| 02 | 1/8 | .218 | .125 | .548 | | .301 | | N/A 3/ | | .363 |
| 03 | 3/16 | .217 | .188 | .548 | | | | | | |
| 04 | 1/4 | .270 | .250 | .579 | | .279 | | | | .341 |
| 05 | 5/16 | .288 | .313 | .610 | | .292 | .062 | | .082 | .354 |
| 06 | 3/8 | .320 | .375 | .592 | | .243 | | | | .305 |
| 07 | 7/16 | .351 | .438 | .623 | | | | | | |
| 08 | 1/2 | .382 | .500 | .654 | | | | | | |
| 09 | 9/16 | .431 | .563 | .752 | | .250 | .109 | .050 | .112 | .359 |
| 10 | 5/8 | .463 | .625 | .783 | | | | | | |
| 11 | 11/16 | .494 | .688 | .814 | | | | | | |
| 12 | 3/4 | .525 | .750 | .845 | | | | | | |
| 13 | 13/16 | .556 | .813 | .877 | | | | | | |
| 14 | 7/8 | .588 | .875 | .908 | | | | | | |
| 15 | 15/16 | .619 | .938 | .939 | | | | | | |
| 16 | 1 | .650 | 1.000 | .970 | .045 ± .005 | | | | | |
| 17 | 1 1/16 | .681 | 1.063 | 1.002 | | | | | | |
| 18 | 1 1/8 | .733 | 1.125 | 1.062 | | .226 | .125 | .070 | .129 | .351 |
| 19 | 1 3/16 | .764 | 1.188 | 1.093 | | | | | | |
| 20 | 1 1/4 | .795 | 1.250 | 1.124 | | | | | | |
| 21 | 1 5/16 | .826 | 1.313 | 1.156 | | | | | | |
| 22 | 1 3/8 | .858 | 1.375 | 1.187 | | | | | | |
| 23 | 1 7/16 | .889 | 1.438 | 1.218 | | | | | | |
| 24 | 1 1/2 | .920 | 1.500 | 1.249 | | | | | | |
| 25 | 1 9/16 | .951 | 1.563 | 1.281 | | | | | | |
| 26 | 1 5/8 | .983 | 1.625 | 1.312 | | | | | | |
| 27 | 1 11/16 | 1.014 | 1.688 | 1.344 | | | | | | |
| 28 | 1 3/4 | 1.045 | 1.750 | 1.374 | | | | | | |
| 29 | 1 13/16 | 1.076 | 1.813 | 1.406 | | | | | | |
| 30 | 1 7/8 | 1.108 | 1.875 | 1.437 | | | | | | |
| 31 | 1 15/16 | 1.139 | 1.938 | 1.468 | .070 ± .015 | | | | | |
| 32 | 2 | 1.170 | 2.000 | 1.499 | | | | | | |
| 33 | 2 1/16 | 1.201 | 2.062 | 1.531 | | | | | | |
| 34 | 2 1/8 | 1.233 | 2.125 | 1.562 | | | | | | |
| 35 | 2 3/16 | 1.264 | 2.188 | 1.594 | | | | | | |
| 36 | 2 1/4 | 1.295 | 2.250 | 1.624 | | | | | | |
| 37 | 2 5/16 | 1.326 | 2.312 | 1.655 | | | | | | |
| 38 | 2 3/8 | 1.358 | 2.375 | 1.687 | | | | | | |



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TABLE I. Clamp dimensions – Continued.

| Dash No. | Harness Nominal O.D. 5/ | C REF | ØD 1/ 5/ | E 2/ 6/ ± .032 | G | J 4/ REF | R ± .010 | S ± .010 | T ± .010 | Y 4/ REF |
|---|---|---|---|---|---|---|---|---|---|---|
| 39 | 2 7/16 | 1.389 | 2.438 | 1.719 | .070 ± .015 | .226 | .125 | .070 | .129 | .351 |
| 40 | 2 1/2 | 1.420 | 2.500 | 1.752 | | | | | | |
| 41 | 2 9/16 | 1.451 | 2.563 | 1.781 | | | | | | |
| 42 | 2 5/8 | 1.483 | 2.625 | 1.812 | | | | | | |
| 43 | 2 11/16 | 1.514 | 2.688 | 1.844 | | | | | | |
| 44 | 2 3/4 | 1.545 | 2.750 | 1.875 | | | | | | |
| 45 | 2 13/16 | 1.576 | 2.812 | 1.906 | | | | | | |
| 46 | 2 7/8 | 1.608 | 2.875 | 1.937 | | | | | | |
| 48 | 3 | 1.670 | 3.000 | 2.000 | | | | | | |
| 50 | 3 1/8 | 1.733 | 3.125 | 2.062 | | | | | | |
| 52 | 3 1/4 | 1.795 | 3.250 | 2.125 | | | | | | |
| 54 | 3 3/8 | 1.858 | 3.375 | 2.187 | | | | | | |
| 56 | 3 1/2 | 1.920 | 3.500 | 2.250 | | | | | | |
| 58 | 3 3/4 | 1.983 | 3.625 | 2.312 | | | | | | |
| 64 | 4 | 2.170 | 4.000 | 2.500 | | | | | | |
| 66 | 4 1/8 | 2.233 | 4.125 | 2.562 | | | | | | |

1/ Diameter "ØD" measured at a 45 degree angle from vertical as shown on figure 1.

2/ Dimension "E" is measured with the clamp installed on a mandrel of “ØD” diameter with a manual tolerance of ±.001.

3/ Clamp sizes 2, 3, and 4 use the grommet shown on figure 1 in lieu of the split cushion.

4/ Reference dimensions “Y” and” J” are provided to gain maximum support for the lower foot (see figure 1) by closely fitting the upper foot (see figure 1) bend radius to the stiffening bead blend radius when closed against each other.

5/ Harness “O.D.” and “ØD” are the nominal diameters for which a clamp size is intended for use

6/ Dimension “E” shall be measured with the clamp installed on a mandrel of “ØD” diameter ± .001, and a spacer with thickness in accordance with dimension “G” between the upper and lower foot as shown on figure 1.

REQUIREMENTS:

1. Unless otherwise specified, dimensions are in inches, tolerances ± .03 for two decimals and ± .010 for three decimals.

2. Wedge (see figure 1) shall be required on sizes 5 through 49. The wedge shall be molded as an integral part of the cushion and contoured to fit harness nominal “O.D.” and “ØD” diameter (see table I).

3. Stiffening bead (see figure 1) shall be required on sizes 2 through 66. All radii on the stiffening bead shall be smooth and blended. No sharp tool marks are allowed.

4. The wedge shall touch the cushion on the lower foot (see figure 1) of clamp when clamp is installed on a mandrel of “ØD” diameter (see table I) ±.001 tolerance, without spacers.



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5. Remove all burrs, scales, sharp edges and break all sharp corners on clamp band. Mold marks and flash are permissible on cushions.

6. Identification shall include part number and manufacturer's name or trademark. Trademark shall be registered in SAE AIR1351. Do not mark in radius. Part number and manufacturer's name or trademark shall be stamped on the clamp loop. Identification may be located on the upper or lower foot when weight reducing hole does not permit marking location shown on figure 1. Part number may be split to accommodate available space.

Part or Identifying Number (PIN) example:

M85052/10-08

- 08 — Appropriate size dash number from table I
- 10 — Number of this specification sheet
- 85052 — Specification number
- M — M Prefix

7. Materials.

a. Metal Band.

2024-0 aluminum alloy per SAE AMS-QQ-A-250/5

Heat treatment - SAE AMS2770, condition T42

Finish - MIL-DTL-5541 Class 1A

b. Cushion. Nitrile butadiene rubber - Color is yellow and the cushion shall be in accordance with MIL-DTL-85052/1.

8. Clamp loop shall include a hinge (see figure 1) to reduce the spring back of the metal band. The spring back force may vary depending on the cushion material.

9. Diametral retention is not required.

EXAMINATION AND TESTS:

1. Locking mechanism.

a. Loop alignment test. The locking mechanism (see figure 1) shall close and lock when mounted without mounting hardware over a mandrel with a diameter "D" within ±.001 inch specified in table II. The mounting holes shall be in alignment within .010 inch. The band width straightness (parallel to the bar axis) shall be within .010 inch except in the area of the stiffening bead.

b. Full locked verification (without mandrel). Position the cushion in accordance with the cushion alignment test. When the clamp is in a fully locked position using mounting hardware on a flat plate, there shall be no gap between the flat portion of the band and the mounting surface. There shall be no visible gap between metal bands extending from the locking area to the far side of the fastener. A gap beyond the radius not extending to the fastener shall be limited to .020 inch maximum. Cushion pinching which causes a gap at the radius is unacceptable. The cushion wedge shall make firm



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rubber to rubber contact sufficient to prevent harness/wire from contacting metal bands.

2. Vibration.

The vibration test shall be conducted using the vibration acceleration limits and break points defined in table III. Simultaneous testing of multiple size clamps and multiple samples is permissible. The configuration of using metal mandrels, mounting method and clamp spacing shown (see figure 3) was done to facilitate multiple configurations testing at the same time. It was also used as a method to establish a standard for testing manufacturers clamp capability within a defined parameter. This may or may not represent an aircraft installation and users must verify their application (see intended use for additional information).

The assembly shall be installed flat on a vibration fixture and shaker table as shown on figure 2. Clamps shall be mounted flat on the vibration fixture with 10-32 UNF-3a bolts (160 KSI or higher strength) and torqued to 30 inch pounds +/- 3 inch pounds. A flat washer shall be used beneath the head of the mounting bolt with the smooth edge against the clamp foot. A torque stripe (indicator) shall be applied to the fasteners to ensure no rotation occurs. A control accelerometer shall be mounted on the vibration fixture. Response accelerometers shall be mounted on the vibration fixture in close proximity to both clamps as shown on figure 3.

Clamp sizes -16, -24, and -32 shall be tested. Two samples of the same size and cushion material shall be mounted on test mandrels with spacing and dimensions as specified on figure 3 and table II. Multiple mandrels with differing types of clamps may be operated on a test fixture at the same time provided they are within the fixture and test equipment capacity, and mandrels do not dynamically interact. If multiple mandrels are tested at one time, at least one mandrel of each type shall have the response accelerometer configuration shown on figure 3.

Vibration testing shall be conducted in the vertical and lateral axis only. After completing one axis of vibration, ne/different clamps shall be used for the second axis.

Test mandrels shall be 6061-T6 aluminum alloy rods in accordance with ASTM B211. All dimensions shall be in accordance with table II with surface roughness of 32 microinches $R_A$ maximum. The mandrels may have shoulders to prevent slippage during vibration. The shoulders shall maintain a positive clearance from the vibration fixture and clamp under test. Shoulder dimension shall not exceed 0.25 inch.



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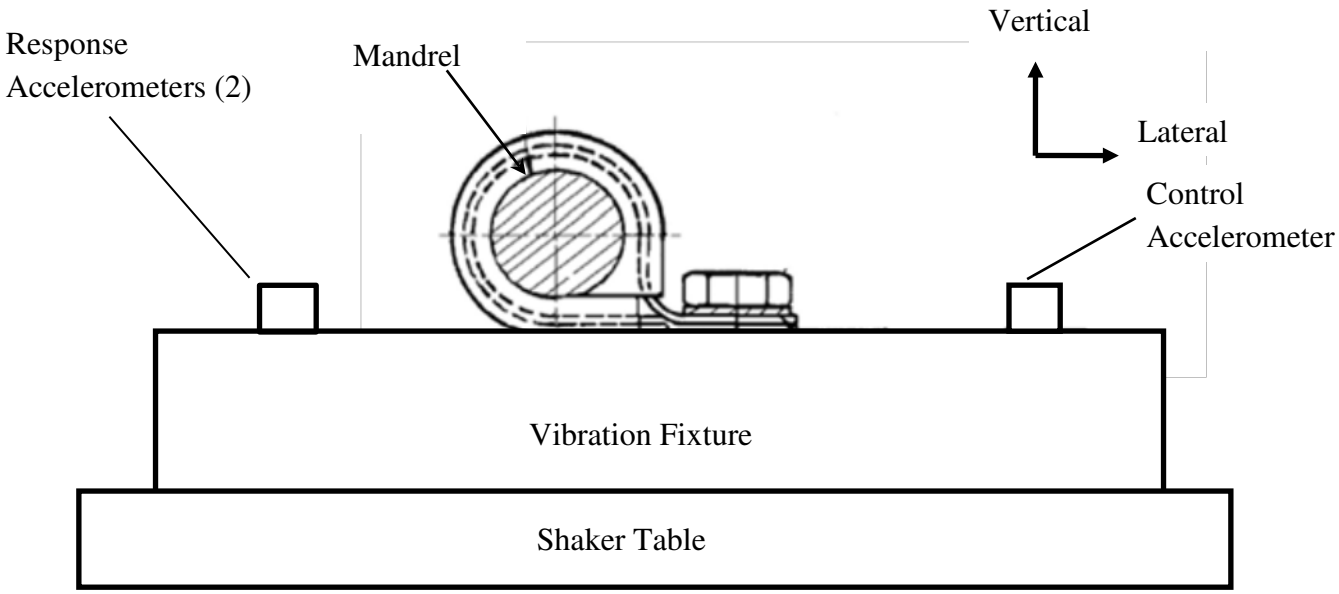



FIGURE 2. Axis definition.



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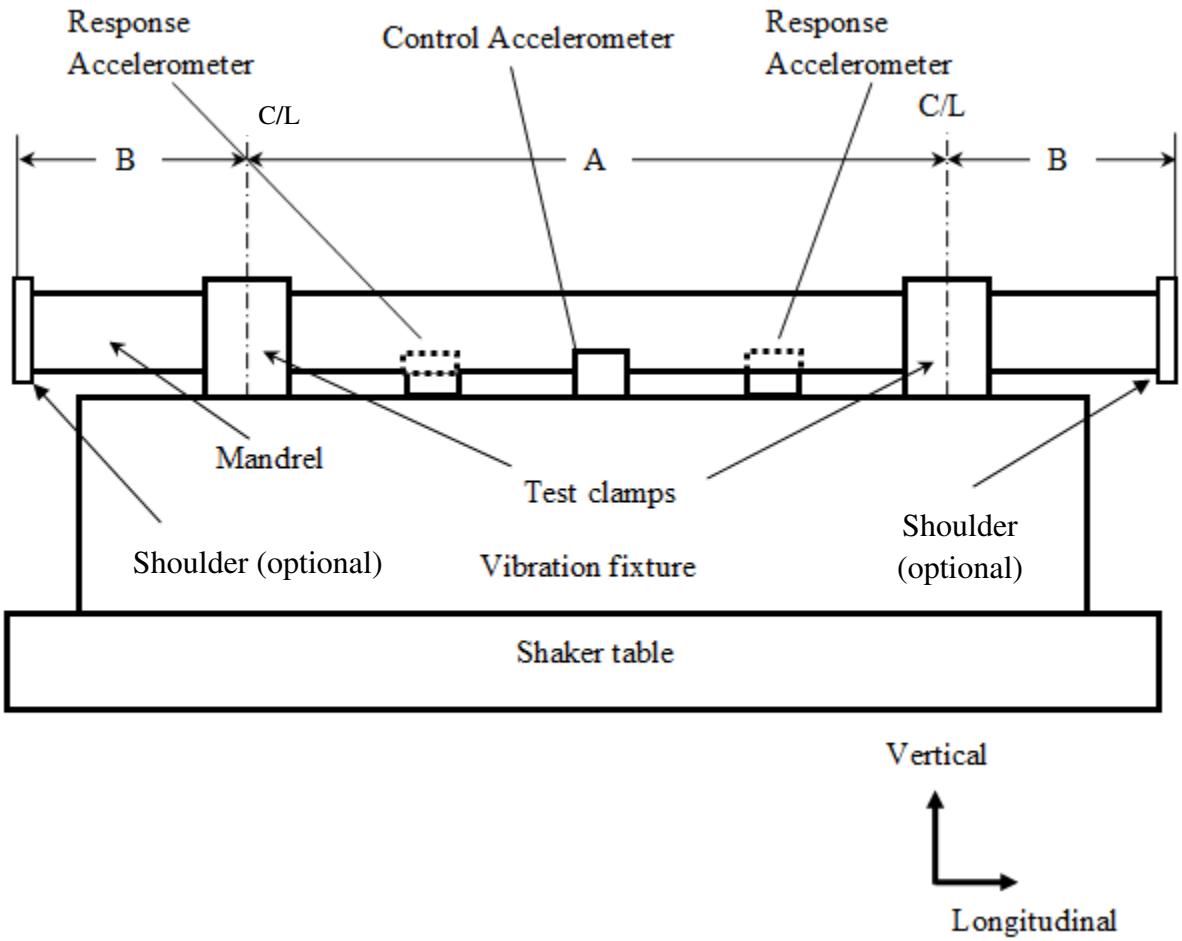



FIGURE 3. Clamp mounting and spacing.



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TABLE II. Mandrel and clamp parameters.

| Clamp size | Mandrel | | Clamp spacing (see figure 3) | |
|---|---|---|---|---|
| | Dia ± .001 [in.] | Length ± .015 [in.] | A [in.] | B[in.] |
| -16 | 1.000 | 7.50 | 6.0 | 0.75 |
| -24 | 1.500 | 7.50 | 6.0 | 0.75 |
| -32 | 2.000 | 7.50 | 6.0 | 0.75 |

TABLE III. Break points for random vibration profile.

| Clamp size | Overall acceleration [$g_{RMS}$]* | 20 Hz Breakpoint [$g^2$/Hz] | 500 Hz Breakpoint [$g^2$/Hz] | 800 Hz Breakpoint [$g^2$/Hz] | Duration [minutes] |
|---|---|---|---|---|---|
| Random vibration vertical | | | | | |
| -16 | 18 | 0.604 | 0.604 | 0.0074 | 17 |
| -24 | 8 | 0.119 | 0.119 | 0.0014 | 45 |
| -32 | 7 | 0.091 | 0.091 | 0.0011 | 24 |
| Random vibration lateral | | | | | |
| -16 | 18 | 0.604 | 0.604 | 0.0074 | 28 |
| -24 | 12 | 0.266 | 0.266 | 0.0030 | 20 |
| -32 | 8 | 0.119 | 0.119 | 0.0014 | 28 |

Clamp failure occurs when visible damage is observed. This includes cracks, distortions, breaks, or releasing of the locking device. Cushion failure occurs when visible damage is observed including: tearing in excess of 0.25 inch, exposure of loop metal, or separation from the clamp.



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3. Retention of qualification. Retention of qualification shall be conducted by the Qualifying Activity and consists of the Examination of product requirements of MIL-DTL-85052 (see 3.4 and 4.4.2). Retention of Qualification shall be performed at 3-year intervals after the initial qualification approval date, unless otherwise notified by the qualifying activity. Initial vibration testing shall be conducted every three retention of qualification cycles (nine years).

4. Maximum weights. Maximum weights are shown in table IV.

TABLE IV. Clamp weights.

| Clamp size | Max weight (pounds) |
|---|---|
| 2-17 | .0272 |
| 18-25 | .0450 |
| 26-33 | .0572 |
| 34-41 | .0702 |
| 42-50 | .0843 |
| 51-58 | .0973 |
| 59-66 | .1099 |

5. Qualification. Testing cannot begin until the supplier has received an authorization letter from the qualifying activity (see Note 2). The qualification process is described in SD-6, "Provisions Governing Qualification." The following examinations or tests shall be conducted by the Qualifying Activity on the clamp assembly: Examination of product, vibration, and thermal shock.

NOTES:

1. Clamp restrictions and applications.

a. Intended use. Clamps specified in this detail specification sheet are intended for the installation of aircraft wire cables, bundles, and harnesses, but may also be used for general clamping of tubing for aircraft systems. Not approved for use on hydraulic tubing. The user must select the appropriate size clamp and spacing between clamps to meet their installation requirements. In addition, the user must select the appropriate temperature rating and fluid exposure capabilities of the cushion material to meet their installation requirements. Not approved for butterfly installation with clamps in use on hydraulic tubing.

These clamps are intended for use as follows:

Temperature range: -65° to +275 °F (for higher temperature rated clamps, see MIL-DTL-85052/8 or /9. Also see MIL-DTL-85052/8 and /9 for fluid exposure limitations.)

Systems**:** All fluid and electrical systems on aircraft and aerospace vehicles using petroleum based hydraulic systems.



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b. Cushion. The nitrile butadiene rubber cushion designator is NW.

c. Wires. Wires must be firmly installed in the clamp (see SAE AS50881 primary support).

d. Deleted.

e. Qualifying activity. The qualifying activity responsible for these clamps is the Naval Air Systems Command, AIR 4.4.5.3, 48298 Shaw Rd., Building 1461, Patuxent River, MD 20670-1900.

2. Test plan. It is recommended that the supplier provide a test plan based on the authorization letter to ensure the supplier and qualifying activity maintain communication and document changes as needed.

## U.S. PATENT NOTICE

The clamps specified in this detail specification sheet are covered by the following United States patents. The Government does not have a royalty-free license.

Patent No. U.S. 7,392,569 and expires on 1 July 2028.

Referenced documents. In addition to MIL-DTL-85052, this specification sheet references:

| | | |
|---|---|---|
| SD-6 | ASTM B211 | SAE AMS-QQ-A-250/5 |
| SAE AIR1351 | SAE AMS2770 | MIL-DTL-5541 |
| SAE AS50881 | MIL-DTL-85052/1 | MIL-DTL-85052/8 |
| MIL-DTL-85052/9 | | |

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## CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99

Preparing activity:
Navy - AS
(Project 5340-2016-012)

Review activities:
Navy - MC, SA, SH
Air Force - 11, 71
DLA - IS

Industry Association:
SAE AE-8C2

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